T3 UV Lagrangian terms grouped by field content

EFT order: 5; loop order: 1. Rows are model configurations and columns are field configurations. The BSM-only UV expression is preferred; the full UV Lagrangian is used as a fallback when the BSM-only conversion is unavailable.

Coupling and interaction notation

symbol	interaction / definition
g_1	$D_\mu \supset i g_1 Y B_\mu$
g_2	$D_\mu \supset i g_2 T^A W_\mu^A$
g_3	$D_\mu \supset i g_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \widetilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$L F S_1 + \text{h.c.}$
y_2	$L F S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\text{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\text{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\text{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$

$C_2(d)$	$(d^2-1)/4$
$T(d)$	$d(d^2-1)/12$
$C^{ij;AB}_{LLS_1S_2}$	$L_iL_jS_1^AS_2^B$
C_{ijab}	$L_iL_j\phi_a\phi_b$
C_5^{ij}	$(L_i^TCL_j)HH$
R_W	$16\pi^2\beta_{C_5} \supset R_W\lambda_5 C_{LLS_1S_2}$

Additional general-interaction couplings

Symbol	Field content	Stored coupling / contraction
λ_7	$(H^\dagger)^2(S_1^\dagger)^2$	`lambdaHHdagS1barS1bar`; invariant contraction model-dependent
λ_{12}	$HH^\dagger S_1^\dagger S_2^\dagger$	`lambdaHHdagS1barS2barCross`; invariant contraction model-dependent
λ_8	$(H^\dagger)^2S_2^2$	`lambdaHHdagS2S2`; invariant contraction model-dependent
λ_{11}	$S_1(S_1^\dagger)^2S_2^\dagger$	`lambdaS1S1bar2S2bar`; invariant contraction model-dependent
λ_9	$(S_1^\dagger)^2(S_2^\dagger)^2$	`lambdaS1bar2S2bar2`; invariant contraction model-dependent
λ_{10}	$S_1^\dagger S_2(S_2^\dagger)^2$	`lambdaS1barS2S2bar2`; invariant contraction model-dependent
λ_{X_1}	$(S_2^\dagger)^4$	`lambdaT3ExtraQ000004I1`; contraction: `((1/3) * (del[SU2L @ S2, #1, #2] * del[SU2L @ S2, #3, #4]) + (1/3) * (del[SU2L @ S2, #1, #3] * del[SU2L @ S2, #2, #4]) + (1/3) * (del[SU2L @ S2, #1, #4] * del[SU2L @ S2, #2, #3]))`
λ_{X_2}	$S_2\,(S_2^\dagger)^3$	`lambdaT3ExtraQ000013I1`; contraction: `((1/3) * (del[SU2L @ S2, #1, #2] * del[SU2L @ S2, #3, #4]) + (1/3) * (del[SU2L @ S2, #1, #3] * del[SU2L @ S2, #2, #4]) + (1/3) * (del[SU2L @ S2, #1, #4] * del[SU2L @ S2, #2, #3]))`
λ_{X_3}	definition unavailable in registry	`lambdaT3ExtraQ000112I1`; symbolic running parameter
λ_{X_4}	definition unavailable in registry	`lambdaT3ExtraQ000202I1`; symbolic running parameter
$\lambda^{(1)}_{X_5}$	$(S_1^\dagger)^2\,S_2\,S_2^\dagger$	`lambdaT3ExtraQ000211I1`; contraction: `del[SU2L @ S2, #3, #4]`
$\lambda^{(2)}_{X_5}$	$(S_1^\dagger)^2\,S_2\,S_2^\dagger$	`lambdaT3ExtraQ000211I2`; contraction: `(del[SU2L @ S2, #1, #2] * del[SU2L @ S2, #3, #4])`
λ_{X_6}	$(S_1^\dagger)^4$	`lambdaT3ExtraQ000400I1`; contraction: `1`
$\lambda^{(1)}_{X_7}$	$S_1\,S_1^\dagger\,(S_2^\dagger)^2$	`lambdaT3ExtraQ001102I1`; contraction: `del[SU2L @ S2, #3, #4]`
$\lambda^{(2)}_{X_7}$	$S_1\,S_1^\dagger\,(S_2^\dagger)^2$	`lambdaT3ExtraQ001102I2`; contraction: `(del[SU2L @ S2, #1, #2] * del[SU2L @ S2, #3, #4])`
λ_{X_8}	definition unavailable in registry	`lambdaT3ExtraQ001201I1`; symbolic running parameter

λ_{X_9}	$S_1\,(S_1^\dagger)^3$	lambdaT3ExtraQ001300I1; contraction: 1
$\lambda_{X_{10}}$	$(H^\dagger)^2\,(S_2^\dagger)^2$	lambdaT3ExtraQ020002I1; contraction: ((1/2) * (eps[SU2L @ fund, #1, #3] * eps[SU2L @ fund, #2, #4]) + (1/2) * (eps[SU2L @ fund, #1, #4] * eps[SU2L @ fund, #2, #3]))
$\lambda_{X_{11}}$	definition unavailable in registry	lambdaT3ExtraQ020020I1; symbolic running parameter
$\lambda_{X_{12}}$	$(H^\dagger)^2\,S_1^\dagger\,S_2^\dagger$	lambdaT3ExtraQ020101I1; contraction: delS2[SU2L, #4, #1, #2]
$\lambda_{X_{13}}$	definition unavailable in registry	lambdaT3ExtraQ020200I1; symbolic running parameter
$\lambda_{X_{14}}$	$(H^\dagger)^2\,S_1\,S_2$	lambdaT3ExtraQ021010I1; contraction: delS2[SU2L, #4, #1, #2]
$\lambda_{X_{15}}$	$(H^\dagger)^2\,(S_1)^2$	lambdaT3ExtraQ022000I1; contraction: ((1/2) * (del[SU2L @ fund, #1, #3] * del[SU2L @ fund, #2, #4]) + (1/2) * (del[SU2L @ fund, #1, #4] * del[SU2L @ fund, #2, #3]))
$\lambda_{X_{16}}$	$H\,H^\dagger\,(S_2^\dagger)^2$	lambdaT3ExtraQ110002I1; contraction: (del[SU2L @ fund, #2, #1] * del[SU2L @ S2, #3, #4])
$\lambda^{(1)}_{X_{17}}$	$H\,H^\dagger\,S_1^\dagger\,S_2^\dagger$	lambdaT3ExtraQ110101I1; contraction: (del[SU2L @ fund, #4, #1] * eps[SU2L @ fund, #2, #3])
$\lambda^{(2)}_{X_{17}}$	$H\,H^\dagger\,S_1^\dagger\,S_2^\dagger$	lambdaT3ExtraQ110101I2; contraction: (del[SU2L @ fund, #4, #1] * eps[SU2L @ fund, #2, #3])
$\lambda_{X_{18}}$	$H\,H^\dagger\,(S_1^\dagger)^2$	lambdaT3ExtraQ110200I1; contraction: del[SU2L @ fund, #2, #1]
λ_{X_1}	$(S_2^\dagger)^4$	lambdaT3GeneralQ000004I1; contraction: ((1/3) * (del[SU2L @ S2, #1, #2] * del[SU2L @ S2, #3, #4]) + (1/3) * (del[SU2L @ S2, #1, #3] * del[SU2L @ S2, #2, #4]) + (1/3) * (del[SU2L @ S2, #1, #4] * del[SU2L @ S2, #2, #3]))
λ_{X_2}	$S_2\,(S_2^\dagger)^3$	lambdaT3GeneralQ000013I1; contraction: ((1/3) * (del[SU2L @ S2, #1, #2] * del[SU2L @ S2, #3, #4]) + (1/3) * (del[SU2L @ S2, #1, #3] * del[SU2L @ S2, #2, #4]) + (1/3) * (del[SU2L @ S2, #1, #4] * del[SU2L @ S2, #2, #3]))
$\lambda^{(1)}_{X_5}$	$(S_1^\dagger)^2\,S_2\,S_2^\dagger$	lambdaT3GeneralQ000211I1; contraction: del[SU2L @ S2, #3, #4]
$\lambda^{(2)}_{X_5}$	$(S_1^\dagger)^2\,S_2\,S_2^\dagger$	lambdaT3GeneralQ000211I2; contraction: ((1/2) * (del[SU2L @ S2, #1, #3] * del[SU2L @ S2, #2, #4]) + (1/2) * (del[SU2L @ S2, #1, #4] * del[SU2L @ S2, #2, #3]))
λ_{X_6}	$(S_1^\dagger)^4$	lambdaT3GeneralQ000400I1; contraction: 1
$\lambda^{(1)}_{X_7}$	$S_1\,S_1^\dagger\,(S_2^\dagger)^2$	lambdaT3GeneralQ001102I1; contraction: del[SU2L @ S2, #3, #4]
$\lambda^{(2)}_{X_7}$	$S_1\,S_1^\dagger\,(S_2^\dagger)^2$	lambdaT3GeneralQ001102I2; contraction: ((1/2) * (del[SU2L @ S2, #1, #3] * del[SU2L @ S2, #2, #4]) + (1/2) * (del[SU2L @ S2, #1, #4] * del[SU2L @ S2, #2, #3]))
λ_{X_9}	$S_1\,(S_1^\dagger)^3$	lambdaT3GeneralQ001300I1; contraction: 1

$\lambda_{X_{10}}$	$(H^\dagger)^2\,(S_2^\dagger)^2$	`lambdaT3GeneralQ020002I1`; contraction: `((1/2) * (eps[SU2L @ fund, #1, #3] * eps[SU2L @ fund, #2, #4]) + (1/2) * (eps[SU2L @ fund, #1, #4] * eps[SU2L @ fund, #2, #3]))`
$\lambda_{X_{12}}$	$(H^\dagger)^2\,S_1^\dagger\,S_2^\dagger$	`lambdaT3GeneralQ020101I1`; contraction: `delS2[SU2L, #4, #1, #2]`
$\lambda_{X_{14}}$	$(H^\dagger)^2\,S_1\,S_2$	`lambdaT3GeneralQ021010I1`; contraction: `delS2[SU2L, #4, #1, #2]`
$\lambda_{X_{15}}$	$(H^\dagger)^2\,(S_1)^2$	`lambdaT3GeneralQ022000I1`; contraction: `((1/2) * (del[SU2L @ fund, #1, #3] * del[SU2L @ fund, #2, #4]) + (1/2) * (del[SU2L @ fund, #1, #4] * del[SU2L @ fund, #2, #3]))`
$\lambda_{X_{16}}$	$H\,H^\dagger\,(S_2^\dagger)^2$	`lambdaT3GeneralQ110002I1`; contraction: `(del[SU2L @ fund, #2, #1] * del[SU2L @ S2, #3, #4])`
$\lambda^{(1)}_{X_{17}}$	$H\,H^\dagger\,S_1^\dagger\,S_2^\dagger$	`lambdaT3GeneralQ110101I1`; contraction: `(del[SU2L @ fund, #4, #1] * eps[SU2L @ fund, #2, #3])`
$\lambda_{X_{18}}$	$H\,H^\dagger\,(S_1^\dagger)^2$	`lambdaT3GeneralQ110200I1`; contraction: `del[SU2L @ fund, #2, #1]`
$\mu_{\text{T3GeneralS1S2}}$	scalar bilinear	`muT3GeneralS1S2`; see native Lagrangian for contraction
$\mu_{\text{T3GeneralS1Sq}}$	scalar bilinear	`muT3GeneralS1Sq`; see native Lagrangian for contraction
$\mu_{\text{T3GeneralS2Sq}}$	scalar bilinear	`muT3GeneralS2Sq`; see native Lagrangian for contraction
y_1^{conj}	$L\,F\,S_1^\dagger$	`y1Conj`; conjugate-scalar Yukawa
y_2^{conj}	$L\,F\,S_2^\dagger$	`y2Conj`; conjugate-scalar Yukawa

Field combinations 1 of 2

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	$S_1\,S_2$
A	0	1	N/A	3	N/A	2	N/A			
cont.										
cont.										
cont.										
cont.										
cont.										
A	−2	1	N/A	3	N/A	2	N/A			
cont.										
cont.										
cont.										
cont.										
cont.										
A	−4	1	N/A	3	N/A	2	N/A			
cont.										
cont.										
cont.										
B	1	2	N/A	2	N/A	1	N/A			
cont.										
cont.										
B	−1	2	N/A	2	N/A	1	N/A			
cont.										
cont.										
cont.										
cont.										

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	$S_1\,S_2$
cont.										
B	-3	2	N/A	2	N/A	1	N/A			
cont.										
cont.										
C	1	2	N/A	2	N/A	3	N/A			
cont.										
cont.										
C	-1	2	N/A	2	N/A	3	N/A			
cont.										
cont.										
cont.										
cont.										
cont.										
C	-3	2	N/A	2	N/A	3	N/A			
cont.										
cont.										
D	0	3	N/A	1	N/A	2	N/A			
cont.										
cont.										
cont.										
cont.										
cont.										
cont.										
D	2	3	N/A	1	N/A	2	N/A			
cont.										
cont.										
cont.										
D	-2	3	N/A	1	N/A	2	N/A			
cont.										
cont.										
cont.									$+\mathrm{Bar}(\mathrm{Coupling}(\lambda_{X_2},\{\},0))\mathrm{Field}(S_2,\mathrm{Scalar},\{\},\{\})^3\mathrm{Bar}(\mathrm{Field}(S_2,\mathrm{Scalar},\{\},\{\}))$	
									$+\mathrm{Bar}(\mathrm{Coupling}(\lambda_{X_4},\{\},0))\mathrm{Field}(S_2,\mathrm{Scalar},\{\},\{\})^4$	
									$+\mathrm{Field}(S_2,\mathrm{Scalar},\{\},\{\mathrm{Index}(\mathrm{d\$\$1},\mathrm{Lorentz})\})\mathrm{Bar}(\mathrm{Field}(S_2,\mathrm{Scalar},\{\},\{\mathrm{Index}(\mathrm{d\$\$1},\mathrm{Lorentz})\}))$	
cont.										
cont.										
E	0	3	N/A	3	N/A	2	N/A			
cont.										
cont.										
cont.										
cont.										
cont.										
cont.										
E	2	3	N/A	3	N/A	2	N/A			
cont.										
cont.										
cont.										
E	-2	3	N/A	3	N/A	2	N/A			
cont.										
cont.										
cont.										
cont.										
cont.										
cont.										
E	-4	3	N/A	3	N/A	2	N/A			
cont.										
cont.										
cont.										

Field combinations 2 of 2

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	constant
A	0	1	N/A	3	N/A	2	N/A	
A	−2	1	N/A	3	N/A	2	N/A	
A	−4	1	N/A	3	N/A	2	N/A	
B	1	2	N/A	2	N/A	1	N/A	
B	−1	2	N/A	2	N/A	1	N/A	
B	−3	2	N/A	2	N/A	1	N/A	
C	1	2	N/A	2	N/A	3	N/A	
C	−1	2	N/A	2	N/A	3	N/A	
C	−3	2	N/A	2	N/A	3	N/A	
D	0	3	N/A	1	N/A	2	N/A	
D	2	3	N/A	1	N/A	2	N/A	
D	−2	3	N/A	1	N/A	2	N/A	
E	0	3	N/A	3	N/A	2	N/A	
E	2	3	N/A	3	N/A	2	N/A	
E	−2	3	N/A	3	N/A	2	N/A	
E	−4	3	N/A	3	N/A	2	N/A	